



The Islamia University of Bahawalpur

TERM PROPOSAL (Research Methods)

A Review on Explainable Identification of Human vs. Large Language Model (LLM) Generated Text

By: Jawwad Ahmad (F25NDOCS3W04013) – MSCS 2nd

Supervisor

Dr. Ahmad Niaz

Head of Department, Software Engineering

EXECUTIVE SUMMARY

The rapid advancement of Large Language Models (LLMs) has significantly transformed the landscape of digital content creation, enabling the generation of highly coherent and human-like text. While these developments offer substantial benefits, they also introduce serious concerns related to academic integrity, misinformation, and the authenticity of online content. As a result, distinguishing between human-written and AI-generated text has become an important research challenge in natural language processing.

Although existing detection approaches—ranging from traditional machine learning techniques to advanced transformer-based models—have demonstrated promising performance, most of these systems operate as “black boxes.” This lack of transparency limits users’ ability to understand and trust the decisions made by such models, especially in sensitive domains like education, journalism, and research evaluation.

This proposal presents a structured plan for a review article that explores the intersection of human vs. LLM-generated text detection and Explainable Artificial Intelligence (XAI). The primary objective is to critically analyze existing detection methods while examining how explainability techniques, such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations), contribute to making these systems more transparent and interpretable.

Specifically, the review aims to (i) categorize existing detection paradigms, including traditional, deep learning, and stylometric approaches, (ii) evaluate the effectiveness of XAI methods in identifying key linguistic and stylistic features, (iii) assess the generalization capability of detection models across different LLMs and domains, and (iv) identify existing research gaps and limitations in current approaches.

By synthesizing recent literature (2023–2026), this study seeks to provide a comprehensive and coherent understanding of explainable AI-based text detection. The findings of this review are expected to contribute toward the development of more reliable, transparent, and trustworthy detection systems, while also offering valuable insights for researchers, educators, and practitioners dealing with AI-generated content.

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1. INTRODUCTION

The evolution of generative AI, particularly Large Language Models (LLMs) like GPT-4, Llama, and Claude, has revolutionized content creation but also introduced complex challenges in distinguishing between human-authored and machine-generated text. As these models become increasingly sophisticated, the “turing test” of text authenticity is becoming harder to pass, necessitating the development of robust detection tools. However, the mere identification of AI-generated text is no longer sufficient in critical domains such as academia, journalism, and legal proceedings. There is a growing demand for explainability—the ability to understand *why* a piece of text is flagged as AI-generated.

Explainability in AI-generated text detection (AGTD) is crucial for building trust and ensuring fairness. Techniques such as LIME and SHAP have emerged as prominent tools for interpreting complex models, allowing researchers to visualize feature importance and token-level contributions (Masih et al., 2025). This proposal advocates for a systematic review of the current state of explainable detection, moving beyond binary classification accuracy to focus on the qualitative insights provided by XAI. The review will explore the synergy between traditional stylometric analysis and modern transformer-based detectors, as highlighted in the HULLMI framework (Joshi et al., 2024).

2. BACKGROUND/LITERATURE REVIEW

Preliminary research indicates a clear division in methodologies. Early approaches relied heavily on stylometry and classical ML. For instance, (Aditya Shah, 2023) utilized stylistic features like syllable count and punctuation ratios, using LIME and SHAP to show how indices like Simpson’s Index drive classification. Similarly, (McGovern et al., 2024) explored “fingerprints” through POS tag distributions and frequent tokens, though they focused more on visualization than formal XAI.

The advent of transformers shifted the focus toward deep learning. The HULLMI framework (Joshi et al., 2024) demonstrates that while traditional models like Random Forest and Naive Bayes remain relevant, transformer-based detectors like RoBERTa-Sentinel offer superior performance when paired with LIME for transparency. Other studies have integrated SHAP with DistilBERT to assign feature importance values to specific words, uncovering the reasoning behind detecting short ChatGPT-generated reviews (Mitrović et al., 2023).

Recent work has also moved toward multi-label classification and attribution. (Najjar et al., 2025) have leveraged XAI to differentiate between multiple LLMs (Llama, Bard, Claude), showing that feature importance profiles can create “author profiles” for different AI sources. This suggests that explainability is not just a secondary feature but a core component of advanced text attribution frameworks.

LITERATURE REVIEW COMPARATIVE TABLE

Author(s) & Year	Title of the Study	Method Used	Dataset	Explainability Method	Key Findings / Contributions	Limitations
Aditya Shah et al. (2023)	Detecting and Unmasking AI-Generated Texts through Explainable AI using Stylistic Features	Stylometry + Traditional ML (SVM, Logistic Regression)	Custom dataset of human and AI-generated texts	LIME, SHAP	Used stylistic features (syllable count, punctuation ratios, Simpson's Index) to classify text. LIME and SHAP revealed which stylistic indices most drive classification decisions.	Limited dataset diversity; focused mainly on English text.
Mitrovic et al. (2023)	ChatGPT or Human? Detect and Explain. Explaining Decisions of ML Models for Detecting Short ChatGPT-generated Text	Deep Learning (DistilBERT)	Short ChatGPT-generated text reviews vs. human reviews	SHAP	SHAP integrated with DistilBERT to assign feature importance to specific tokens, uncovering the reasoning behind classification of short ChatGPT-generated reviews.	Limited to short texts; may not generalize to longer documents.
Adilazuarda et al. (2023)	Beyond Turing: A Comparative Analysis of Approaches for Detecting Machine-Generated Text	Comparative study (ML, Transformers, Stylometry)	Multiple benchmark corpora including multilingual datasets	Not explicitly specified	Comprehensive comparison of detection paradigms; highlights trade-offs between accuracy and interpretability. Noted that no single method dominates across all domains.	Limited focus on explainability; multilingual coverage is partial.
McGovern et al. (2024)	Your Large Language Models Are Leaving Fingerprints	Stylometry (POS tag distributions, frequent tokens)	Texts generated by multiple LLMs (GPT, Llama, etc.)	Attention visualization (informal)	LLMs leave consistent stylistic fingerprints through POS distributions and token frequencies. Focused more on visualization than formal XAI; results suggest model-specific linguistic patterns.	No formal XAI (LIME/SHAP); primarily exploratory visualization.
Joshi et al. (2024)	HULLMI: Human vs LLM Identification with Explainability	Transformer-based (RoBERTa-Sentinel) + Traditional ML (Random Forest, Naive Bayes)	HULLMI benchmark dataset (multi-source, multi-LLM)	LIME	Foundational framework combining traditional and deep learning detectors with LIME for transparency. RoBERTa-Sentinel achieved superior performance while LIME provided interpretable explanations for model decisions.	Primarily English-language; limited adversarial robustness testing.
Masih et al. (2025)	Classifying Human vs. AI Text with Machine Learning and Explainable Transformer Models	Transformer-based + Traditional ML	Mixed human and AI-generated text corpus	LIME, SHAP	Demonstrated that explainability techniques can visualize feature importance and token-level contributions. Showed how LIME and SHAP demystify opaque model decisions in AI text classification.	Dataset scope may limit cross-domain generalizability.

Author(s) & Year	Title of the Study	Method Used	Dataset	Explainability Method	Key Findings / Contributions	Limitations
Baharifar et al. (2025)	Interpretable AI for Classifying Human- and LLM-Generated Medical Misinformation with Multi-Modal Features	Multi-modal ML (text + other features)	Medical misinformation dataset	Interpretable AI / Feature attribution	Applied interpretable AI for domain-specific (medical) detection of LLM-generated misinformation. Multi-modal features improved accuracy while explainability increased stakeholder trust.	Domain-specific; limited generalizability to non-medical content.
Iqbal & Kashem (2025)	A Machine Learning Framework for Identifying Sources of AI-Generated Text	Ensemble ML	Multi-source AI-generated and human-written text	Feature importance (ML-based)	Proposed a framework capable of source attribution (identifying which LLM generated text) beyond binary classification. Demonstrated practical applicability across multiple LLM sources.	Scalability to newer LLMs not fully explored.
Mohammadi et al. (2025)	Explainability-Based Token Replacement on LLM-Generated Text	Transformer-based with adversarial analysis	LLM-generated text datasets	LIME, SHAP (token-level)	Novel approach using XAI explanations to guide token replacement, demonstrating vulnerability of detectors to explainability-informed adversarial attacks.	Highlights a security gap; practical defence mechanisms not yet proposed.
Najjar et al. (2025)	Leveraging Explainable AI for LLM Text Attribution: Differentiating Human-Written and Multiple LLM-Generated Text	Multi-label classification (Transformers + Ensemble)	Texts from multiple LLMs (Llama, Bard/Gemini, Claude, GPT) and humans	XAI-based feature importance profiles	XAI used to create distinct 'author profiles' for different LLMs. Feature importance patterns can differentiate between multiple AI sources, moving beyond binary to multi-label attribution.	May require frequent retraining as LLMs evolve.
Przystalski et al. (2025)	Stylometry Recognizes Human and LLM-Generated Texts in Short Samples	Stylometry	Short-text samples from humans and LLMs	SHAP (feature attribution)	Stylometry remains effective even in short-sample scenarios. SHAP revealed specific linguistic markers (grammatical standardization, word frequencies) that LLMs over-utilize as discriminative features.	Short-sample focus; less effective on longer, more varied text.
Zahraa J. Mohammed Ali et al. (2025)	Interpretable Multi-Label Classification of Human- and LLM-Generated Texts Using Transformer Embeddings and XAI	Transformer embeddings (multi-label classifier)	Multi-label human and LLM-generated text dataset	LIME, SHAP	Extended detection to multi-label classification covering multiple LLMs. Transformer embeddings combined with LIME/SHAP provided granular interpretability, showing which textual features distinguish each LLM.	Computationally intensive; scalability may be a challenge.

3. PROBLEM STATEMENT

Current methodologies for detecting LLM-generated text suffer from two primary limitations. First, while transformer-based detectors (e.g., RoBERTa, DistilBERT) achieve state-of-the-art accuracy, their internal decision-making processes are opaque, making it difficult for end-users to trust or verify their outputs (Mitrović et al., 2023). Second, there is a lack of consensus on which linguistic features (e.g., perplexity, burstiness, stylistic markers) are most indicative of AI generation across different models and domains (McGovern et al., 2024; Przystalski et al., 2025).

Without adequate explainability, detection tools may be prone to biases or false positives, particularly when dealing with non-native English speakers or highly technical content. The absence of a unified framework that integrates high-performance detection with intuitive, human-understandable explanations hinders the deployment of these tools in sensitive environments. This research addresses this gap by proposing a review that evaluates how XAI can be integrated into the detection pipeline to provide both accuracy and interpretability.

4. OBJECTIVES

The primary objective of this proposed review is to provide a comprehensive evaluation of the current landscape of explainable human vs. LLM text identification. Specific objectives include:

1. **Categorize Detection Paradigms:** Systematically classify existing detection methods into traditional machine learning (e.g., Random Forest, XGBoost), deep learning (e.g., Transformers), and stylometric approaches (Adilazuarda et al., 2023; Joshi et al., 2024).
2. **Evaluate XAI Integration:** Analyze the application of explainability techniques—specifically LIME and SHAP—in the context of text detection and their effectiveness in identifying discriminative linguistic features (Aditya Shah, 2023; Zahraa J. Mohammed Ali 2025).
3. **Assess Multi-Modal and Multi-LLM Generalization:** Examine how detection models perform across various LLMs (e.g., GPT-3.5, GPT-4, Llama, Bard) and different text types (e.g., academic abstracts, social media, medical misinformation) (Baharifar et al., 2025; Iqbal & Kashem, 2025).
4. **Identify Research Gaps:** Pinpoint limitations in current explainable detection methods, such as susceptibility to adversarial attacks or lack of robustness in short-sample scenarios (Mohammadi et al., 2025; Przystalski et al., 2025).

5. RESEARCH QUESTIONS

To guide the review, the following research questions have been formulated:

RQ1: How do traditional machine learning models compare with transformer-based architectures in terms of detection accuracy and inherent interpretability?

RQ2: To what extent do post-hoc explainability techniques (LIME, SHAP) accurately reflect the linguistic “fingerprints” left by different LLMs?

RQ3: Which stylistic and structural features are consistently identified by XAI tools as the strongest indicators of AI-generated text across diverse datasets?

RQ4: How effective are explainable detection frameworks in identifying sources of text (attribution) vs. simple binary classification (human vs. AI)?

6. SCOPE

The proposed review will focus on scholarly articles published between 2023 and 2026, capturing the most recent advancements in LLM technology and detection strategies. The scope includes:

Models: LLMs such as GPT series, Llama, Claude, Google Gemini (Bard), and specialized models for research or medical text.

Techniques: Fine-tuned transformers, ensemble models, stylistic feature extraction, and fingerprinting methods.

Explainability: Local and global interpretability methods (LIME, SHAP), attention mechanism visualization, and feature attribution.

Domains: Academic writing, online reviews, medical information, and general-purpose corpora like HC3 or M4.

The review will exclude non-textual AI generation (e.g., image or video detection) and will primarily focus on English-language texts, though multilingual studies will be included if they offer unique explainability insights (Adilazuarda et al., 2023).

7. METHODOLOGY FOR THE REVIEW

The review will follow a systematic literature review (SLR) methodology to ensure transparency and reproducibility:

1. **Search Strategy:** Electronic databases including Google Scholar, SciSpace, ArXiv, and IEEE Xplore will be searched using keywords such as “LLM text detection,” “Explainable AI,” “LIME/SHAP for text,” and “human vs AI classification.”
2. **Selection Criteria:** Papers will be screened based on their relevance to the core topic, focusing on those that provide a methodology for both detection and

explanation. High priority will be given to papers that introduce novel datasets or benchmarking frameworks like HULLMI (Joshi et al., 2024).

3. **Data Extraction:** For each selected paper, data will be extracted regarding the detection architecture, the explainability tool used, the datasets employed, and the primary linguistic findings.
4. **Synthesis and Analysis:** A comparative analysis will be conducted to identify trends (e.g., the shift from hand-crafted features to transformer embeddings) and commonalities in explainability results.
5. **Quality Assessment:** Each source will be evaluated based on its experimental rigor, sample size, and the clarity of its XAI application.

8. SIGNIFICANCE

This research proposal is significant for several reasons. For the academic community, it provides a centralized synthesis of the rapidly evolving AGTD field, highlighting the transition toward “Glass-Box” models. For educators and publishers, the focus on explainability offers a pathway toward more justifiable and transparent plagiarism detection tools. Furthermore, by identifying the specific linguistic markers that LLMs over-utilize—such as grammatical standardization or specific word frequencies identified by SHAP (Przystalski et al., 2025)—this review can inform the development of more resilient detection systems. Ultimately, the proposal aligns with the MS research methods curriculum by demonstrating a rigorous application of systematic review techniques to a high-impact, contemporary technological challenge.

9. CONCLUSION

The identification of LLM-generated text is a critical task in the age of generative AI, but it must be accompanied by robust explainability to be truly effective. This proposal has outlined a structured plan for a review article that will evaluate the state of the art in explainable detection. By synthesizing the findings of 10+ core scholarly articles, the resulting review will provide a definitive guide to how XAI techniques like LIME and SHAP are transforming a once-opaque detection process into a transparent and scientifically grounded discipline.

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